



## A Dual-Branch Fusion CNN for Multi-Label RF Signal Classification

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**Try the live console**

Same AI, on your  
phone



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Group NEXA · Universiti Teknologi PETRONAS  
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• SEDIC 2026 · TRACK I · PROJECT OVERWATCH: COGNITIVE CEMA



## A Dual-Branch Fusion CNN for Multi-Label RF Signal Classification

Who is transmitting, and are they hostile? NEXA reads raw radio samples, names every signal in the window (civilian comms, radar, frequency hopping, jamming) and tells the operator which tier it belongs to. It runs offline, in a browser, with a human in the loop.

Open the live console

Read the brochure

**8**  
signal classes,  
judged  
independently

**128,400**  
training and test  
windows

**3 / 3**  
judged classes  
above the 80%  
bar

**0 servers**  
inference runs in  
the browser

## 02 THE MISSION

## One band. Friends, the military and an adversary, all at once.

In contested spectrum, civilian traffic, military emissions and deliberate interference overlap in time and frequency. A detector that can only name one signal per window misses the most important case: a jammer sitting on top of a real transmission. NEXA answers yes or no for each class separately, so it can report both.

**CIVILIAN**

Ordinary digital communications

BPSK QPSK  
16QAM 64QAM

**MILITARY / CEMA**

High priority emitters

LFM radar FHSS

**HOSTILE**

Deliberate interference

Barrage Tone  
Sweep

**EMPTY CHANNEL**

Nobody transmitting

Noise floor

One window, more than one answer: { QPSK, JAMMING }

a civilian link being jammed, reported as both.

## 03 HOW IT WORKS

## From raw samples to a threat verdict in five steps.

Every window is 512 IQ samples at 3.2 MHz: 160 microseconds of radio. The same format is enforced for every source, so the model learns the signal and not where it came from.

01

**Capture**

Raw IQ from a receiver, a recording or a SigMF file, cut into windows.

$2 \times 512 \cdot 3.2 \text{ MHz}$

02

**Signal processing**

Normalise power, then build a spectrogram (STFT) alongside the raw samples.

$\text{IQ} + \text{time-frequency}$

03

**Two-branch AI**

One branch reads phase in the raw IQ; the other reads chirps and hops in the spectrogram. They are fused.

dilated 1D-CNN || 2D-CNN

04

**Eight decisions**

A separate calibrated yes/no per class, so overlapping signals are all reported.

$8 \times \text{sigmoid} \cdot 5\text{-model ensemble}$

05

**Verdict**

Detections are grouped into events on a timeline and ranked Hostile, Military, Civilian.

$\text{tier} + \text{event log}$

**Why two branches?**

Civilian modulations are defined by phase, which only raw IQ keeps. Radar chirps and frequency hops are clearest in a spectrogram. Neither view does both jobs, so the model uses both on the same window.

**Where the data comes from**

Real benchmark data where it exists (RadioML 2018.01A, RadChar), our own generators where it does not (FHSS, jamming, empty channel), plus overlays and mixtures of up to three signals.

## 07 THE TEAM

## Group NEXA Universiti Teknologi PETRONAS.



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Jamming · inference and packaging



Try the live console  
Same AI, on your phone



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## What NEXA brings to contested spectrum.

Seeing who is transmitting is only useful if it leads to a faster, better decision. NEXA makes that call offline on one laptop, learns from its operators' corrections, and shows the evidence behind every figure.

## OPERATIONS

### Clearer calls, sooner.

#### Threats surface first

Jamming is ranked ahead of routine traffic, so attention goes where the risk is.

#### Works disconnected

The full five-model ensemble runs on one laptop, with no cloud or server to depend on.

#### Overlap is reported

A jammer on a live link is reported as both signals. Overlapping jamming is still found 84 to 94% of the time from -6 dB up.

#### Decisions on record

Two-person approval and a tamper-evident log stand behind every correction.

## DEPLOYMENT

### Light to field, simple to run.

#### Runs where a browser runs

Standard ONNX models and static hosting; no GPU needed to classify.

#### Improves in service

Operator corrections feed guarded retraining, with every version kept for rollback.

#### Fits existing workflows

SigMF recordings in and out, with PDF, Excel, CSV and JSON reports.

#### Small enough to field

About 182,000 parameters per model, so the five-model ensemble runs on an ordinary laptop.

## RESEARCH

### A method that can be tested.

#### Evidence, labelled

Each figure is marked as a requirement, a measurement, a diagnostic or an interpretation.

#### Clean evaluation

Thresholds are set on validation data only; every figure comes from 19,260 held-out test windows.

#### Honest at low SNR

Recall is reported per class and per noise level, not as one average.

#### Reproducible

Versioned configurations, fixed seeds and an automated test suite.

## Above the bar on every judged class.

Recall on 19,260 held-out test windows, 5-model ensemble, thresholds calibrated on validation data only. The black tick on each meter is the organiser's 80% pass mark.

## MILITARY

## LFM RADAR

# 84.2%

PASS · +4.2 PTS

## MILITARY

## FREQUENCY HOPPING (FHSS)

# 82.9%

PASS · +2.9 PTS

## HOSTILE

## JAMMING

# 84.0%

PASS · +4.0 PTS

## 97.0%

 comms vs jamming  
 discrimination, 16,290 windows

## 99.7%

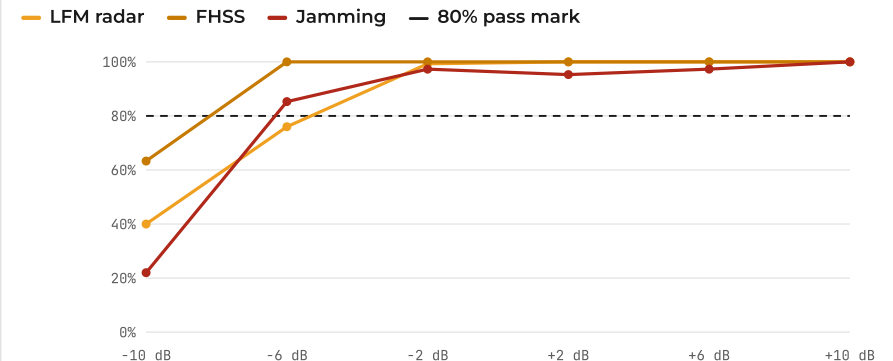
 jamming precision: when it says  
 jammed, it almost always is

## 92.0%

 correct tier (Civilian, Military,  
 Hostile, Empty)

### Recall against noise level, judged classes

Single-signal test windows. Left is noisier: at -10 dB the signal sits below the noise.

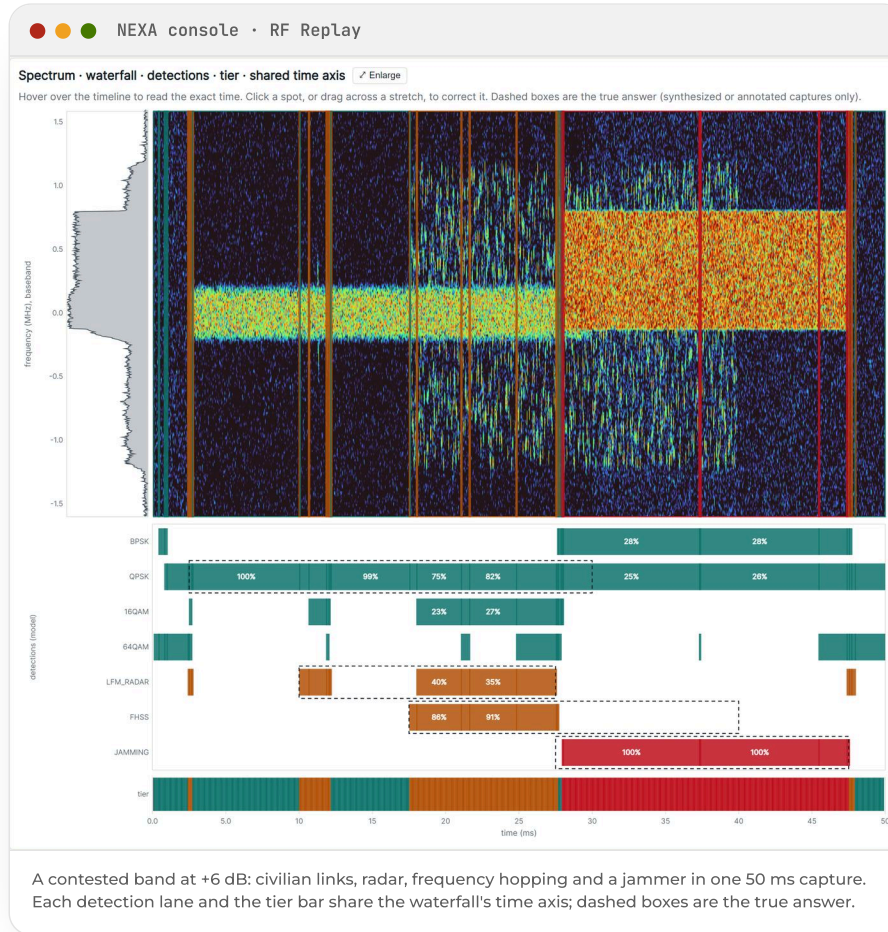


### What we say before you ask

- The pass is an average over the test set. At -10 dB all three judged classes fall below 80%.
- Radar and FHSS are tuned for recall: about 6 in 10 of their alarms are correct. A missed emitter costs more than a second look.
- The drop is confined to the noisiest conditions. From -2 dB upward, radar, FHSS and jamming are each detected at 95% or better on single-signal windows.

# Not just a model. A console an operator can use.

NEXA ships as a complete workstation: live view, analysis, review, reporting and retraining, with roles and an audit trail. Every screen on this page is the working console, captured as it ran.



## LIVE DEMO

## See NEXA run

A capture goes in, the five-model ensemble classifies it in the browser, and the operator reviews the events. No server, no internet.

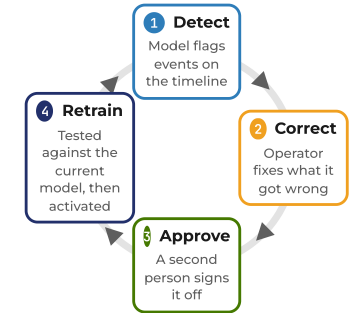
[Open the live console](#)

## RETRAINING THAT CANNOT GET WORSE

## The learning loop

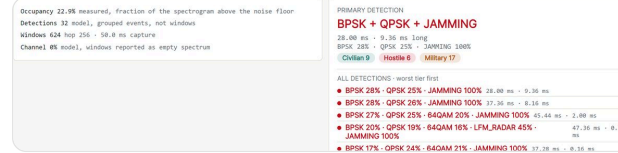
NEXA learns from the operators who use it. What they correct becomes training data, but only after a second person approves it. Approved corrections fine-tune the model, mixed with the original training data so it does not forget what it already knew.

- A retrained version is offered only if it does no worse on radar, FHSS and jamming.
- Only a second person can activate it.
- Every version is kept, so any of them can be brought back.

[Page: Model](#)


## RF Replay: live spectrum, waterfall and detections

Connect a receiver or replay a file as if it were live. Spectrum, waterfall, detections and the threat tier share one time axis, updated dwell by dwell. The verdict panel puts the worst tier first.


[Page: RF Replay](#)


## See what the model sees

Open any 160  $\mu$ s window: measured amplitude next to the model's own attention, plus a civilian constellation showing raw IQ against recovered symbols.


[Page: Signal Analysis](#)


## Assistant that cannot invent numbers

Ask in plain language. Answers come from the stored analyses; an optional local language model only rephrases the question, so it never sees or makes up data.


[Page: Assistant](#)

## GOVERNANCE &amp; ADMINISTRATION

## Accountable by design

## Role-based access

Operator, Analyst and Admin accounts, each with its own permissions. Every account change is logged.

## Tamper-evident audit trail

Every analysis, export and deletion is logged, each entry chained to the one before, so one click proves nothing was edited.

## Offline operation

The models run inside the browser through ONNX, with no internet connection and no inference server.

## Four-eyes review

A correction or a missed signal becomes training data only after a different named person approves it.

## Controlled reporting

PDF, Excel, CSV, JSON, a SigMF bundle or raw IQ with SHA-256 hashes, each carrying a classification banner.